What is Presented and What is Published: Comparing Topic Distributions Across Behavior-Analytic Conferences and Journals

Preprint

David J. Cox*
Institute for Applied Behavioral Science, Endicott College, Beverly, MA, USA
Mosaic Pediatric Therapy, Charlotte, NC, USA

Abstract

Conferences and peer-reviewed journals are the primary outlets where scientists and practitioners disseminate new findings and communicate about their work. For academics, publications are the professional currency and, thus, contingencies may favor presenting their work at conferences and publishing in peer-reviewed journals. In contrast, the contingencies surrounding service delivery and professional practice may support conference presentations but not publication in peer-reviewed journals. The purpose of the present analysis was to determine whether these putative contingencies lead to misalignment between topics presented and published. To do so, we embedded the titles of 5,327 presentations from three behavior-analytic conferences and 6,872 articles from fourteen behavior-analytic journals published from 2016 through 2026. Of the 12,199 titles, 1,552 (12.7%), including 19.7% of all presentations, sat among titles that were in a reliably presentation-heavy neighborhood and had fewer than 10 closely related articles. Presentation-heavy titles concentrated in work on professional development, collaboration between research and practice, the profession itself, clinical service delivery, and school systems; and with several smaller concentrations in some basic and translational areas. Importantly, this analysis does not account for all conference presentations or all published journal articles during the same time period. Thus, the current analysis is best considered as a starting point for discussions about: (a) potential differences between presented and published topics in behavior analysis; (b) differential contingencies surrounding presenting and publishing among researchers and service delivery professionals; and (c) how to promote dissemination of rigorous, data-based advances spanning research and practice in behavior analysis.

Keywords behavior analysis · conference presentations · continuing education · publication · semantic embedding · verbal community

Behavior analysts communicate with one another about their work in at least two settings that leave a permanent product. They present at conferences and they publish in journals. Each permanent product provides a sample of what people communicating in those outlets attend to, find important, and believe is worth sharing with other behavior analysts. Given the volume and function of conference presentations and peer-reviewed publications in behavior analysis, neither likely contains a complete record of the then-current state of the field. For example, conferences may favor work that is timely, practice-facing, or may not yet be ready to submit for a peer-reviewed publication. In contrast, journals favor work that has survived peer review and aligns with the empirical or argumentative standards that academic journal boards have been shaped to uphold (e.g., Critchfield & Reed, 2017; Glenn, 1988; Shadish et al., 2016).

Alignment between topics presented and published is now more important than it was historically for practicing behavior analysts certified by the Behavior Analyst Certification Board (BACB). Specifically, Board Certified Behavior Analysts (BCBAs) and doctoral-level BCBAs (BCBA-Ds) must acquire 32 continuing education units (CEUs) every two years to maintain their certification (BACB, 2025). Much of that continuing education is delivered at conferences. And, effective July 1, 2026, the BACB (2026) requires that continuing education instructors "have demonstrable expertise in the

*Address correspondence to cox.david.j@gmail.com. Data and code: https://github.com/david-j-cox/verbal-communities-observatory.

subject matter being presented" which "must include one of the following": a track record of published peer-reviewed research or books on the subject; at least five years of practical experience with the subject; at least three years of experience teaching, training, or mentoring on the subject; or a combination of these. Publication is, thus, one route among several whereby a CEU presenter can demonstrate their expertise.

Recently, several researchers have analyzed the related publication records of speakers delivering CEU content. For example, Ayres Boggs et al. (2026) evaluated 735 CEU presentations given by keynote and invited speakers at state or regional ABA conferences across three years. Using each speaker's count of peer-reviewed publications on the presentation topic (hereafter, topic-specific publications), 31% of speakers had zero topic-specific publications, and the percentage of speakers with zero topic-specific publications increased across the three years. Rapp et al. (2025) extended the analysis to a single recent conference reporting that, of 17 presentations, 14 (82.4%) had a lead speaker with zero topic-specific publications. A replication in Canada found over half of speakers had no topic-specific publications (Mohamed et al., 2026).

Two ethical concerns arise from this tension (Rapp et al., 2025). First, how might one ethically justify approving CEUs on content delivered by individuals who lack scientific expertise in the subject matter? Second, how might one ethically justify inviting speakers to present on a topic with little peer-reviewed behavior-analytic literature? The first question is about speakers and conference organizers. The second question is about topic alignment across conferences and peer-reviewed publications. The present analysis attempts to provide data for the second question. Specifically, what topics are important enough to conference organizers that speakers are sought or selected to educate conference attendees but for which the peer-reviewed published literature is less than expected?

The purpose of this report is therefore twofold: (a) to identify the topics that behavior analysts present at a higher rate than they publish relative to the field's overall ratio; and (b) among those, to identify the topics whose absolute base of published work is small. Ayres Boggs et al. (2026) and Rapp et al. (2025) have taken up the speaker-level question of who is qualified to present. We describe the topic-level distribution underneath presentation-publication alignment and offer these data as a starting point for discussions about research, publication, and professional training.

Method

We sought to compare what behavior analysts present at conferences relative to what they publish in journals within a common recent window. Analysis of textual data typically requires at least three decisions by the researcher (Gentzkow et al., 2019; Grimmer & Stewart, 2013). First, identify the set of text that will be used for the analysis (i.e., determine the corpora). Second, decide how the textual data will be converted into numbers so that they can be analyzed (e.g., embed the data). Finally, decide how the numerically-represented textual data will be used to answer the question(s) of interest (i.e., analytics and modeling). For those new to these methods, an interactive webpage containing visualizations and demonstrations of the entire process can be found at: https://david-j-cox.github.io/verbal-communities-observatory/.

Corpora

The *presented corpus* pooled three conference sources, all reduced to presentation titles for cross-source comparability. These included: the annual conventions of the Association for Behavior Analysis International (ABAI; 2016-2026; n = 4,443), the annual conventions of the Association of Professional Behavior Analysts (APBA; 2023-2026; n = 252), and the annual meeting programs of the Society for the Quantitative Analyses of Behavior (SQAB; 2016-2026; n = 632). These three sources were chosen deliberately as they are unlike one another and to avoid the results favoring one area of behavior analysis (i.e., basic, applied, practice). Presentation titles were obtained for a total of 5,327 conference presentations.

Two features of the *presented* dataset are worth making explicit. First, conference sources differed in the kind of events they record and made available. Seventy-eight distinct event labels were identified for ABAI records (e.g., symposia, paper sessions, panel discussions, invited presentations, workshops, B. F. Skinner Lecture Series events). Invited events of all kinds (invited presentations, Skinner Lecture Series events, tutorials, and workshops) accounted for 648 of the 4,443 ABAI records (14.6%). The *presented* dataset is, thus, dominated by ordinary symposia and paper sessions rather than by invited or keynote talks. Second, we counted every presentation regardless of whether it offered continuing education credit. The question we were interested in is what behavior analysts present on rather than being specific to CEU events for certified behavior analysts.

The *published corpus* comprised fourteen behavior-analysis journals spanning 2016-2026 retrieved through OpenAlex. These included: *The Analysis of Verbal Behavior*, *Behavior Analysis in Practice*, *Behavior Analysis: Research and Practice*, *The Behavior Analyst/Perspectives on Behavior Science*, *Behavior and Social Issues*, *Behavior Modification*,

Behavioral Interventions, *Education and Treatment of Children*, the *Journal of Applied Behavior Analysis*, the *Journal of Behavioral Education*, the *Journal of Contextual Behavioral Science*, the *Journal of Organizational Behavior Management*, the *Journal of the Experimental Analysis of Behavior*, and *The Psychological Record*. Article titles were obtained for 6,872 articles. Non-research items (e.g., awards, business meetings, social events, journal front matter, supplementary-material deposits, and memorials or tributes to individuals) were removed from the presentation and published datasets using an anchored text filter.

Numeric Transformation of Textual Data

Each conference presentation and journal article title was converted to a list of numbers (i.e., a vector) that quantitatively describe that title using a pretrained sentence-transformer model (all-MiniLM-L6-v2; Reimers & Gurevych, 2019; Wang et al., 2020). A sentence-transformer is a model that maps a piece of text to a fixed-length vector (384 values for the all-MiniLM-L6-v2 model) such that texts with similar meanings land near one another in that space (Reimers & Gurevych, 2019). Applying the model to titles places presentations and articles on similar subjects near one another regardless of the exact words used (sometimes referred to as *semantic embedding*; e.g., Mikolov et al., 2013; Reimers & Gurevych, 2019). See step 2 of the interactive supplement for a visualization of text embedding.

Transforming textual data using pre-trained models is relatively straightforward (Table 1 shows the pseudocode). Using nontechnical language, a model is a function—a mathematical expression that takes data as an input and returns an output. For example, 2x+3 is a mathematical function that takes a number as an input (entered into the *x* location), multiplies it by two, and then adds three to return an output. To embed the titles, we first downloaded the all-MiniLM-L6-v2 (revision 1110a24) model onto our computer[2]. The all-MiniLM-L6-v2 takes a string of text as its input and returns a list of 384 numbers as its output. Thus, to transform the presentation and article title data to numbers, we simply iterated over our list of titles, passed each title into the model, and received back the list of 384 numbers specific to that title. See step 2 of the interactive supplement for a visualization of the resulting output based on the text input.

The all-MiniLM-L6-v2 was chosen for three reasons. First, it is a general purpose sentence-transformer model trained to place sentences with similar meanings close together. Titles most closely resemble sentences in length and content as opposed to words, paragraphs, or articles where word-embedding, paragraph-embedding, or document-embedding models would have been the appropriate choice, respectively (Mikolov et al., 2013). Second, it is a small model (~22 million parameters) fine-tuned using more than one billion sentence pairs from a six-layer version of MiniLM which was itself distilled from a larger model (Wang et al., 2020). This allowed for sentence embedding of the current corpora in minutes on a laptop. The weights are also open and fixed which allows the vectors to be reproduced to numerical precision. Lastly, it has a 256-word piece input limit which is longer than any presentation or article title. In turn, this ensures that each record could be embedded without being truncated.

Comparing Each Title With Its Neighbors

We can think of the list of numbers as pointing to a specific location in a geometrical space. For example, the upper left panel in Figure 1 shows how a list of two numbers points to a specific location in an *x*-*y* plane. And, the upper right panel in Figure 1 shows how a list of three numbers points to a specific location in an x-y-z space. The lower left panel in Figure 1 shows how we can use the angle between any two data points relative to the origin (cosine similarity), to quantify the similarity between any two points in the graph. In turn, for any one datum, we can obtain a ranked list of the data points most-to-least similar within the graph[3].

Defining Presentation-Heavy Areas. For every title in the corpus, we next sought to determine whether it sat in a part of the embedding space where behavior analysts present more than they publish. In the present corpus, two randomly chosen titles had a median cosine similarity of .18 and the median title's single most similar title had a cosine similarity of .74 (see Appendix A for the corresponding distributional plots). Thus, to determine whether a title was in a

[2]You can interact with many "language models" also via application programming interfaces (APIs). That is, rather than downloading the model to your computer and then passing text into the model to get a numeric output, you can send your text over the Internet to a server where the model is stored. And, the server then returns back to you the list of numbers. APIs are advantageous particularly where the computation needed to store or use the model is significantly slowed by the computer one is working on (e.g., large language models).

[3]Why cosine similarity and not absolute distance? Cosine similarity compares the direction of two vectors and ignores their length. Cosine similarity is commonly used in textual analysis because the length of a text's vector can vary for reasons unrelated to its meaning (e.g., how long or wordy the text is; Manning et al., 2008). Relatedly, the all-MiniLM-L6-v2 model was trained to place texts with similar meanings at small angles from one another (Reimers & Gurevych, 2019) making cosine similarity the best measure to identify similar titles in the present study.

"presentation-heavy" area, we first looked at the 75 titles closest to it using cosine similarity and counted how many were presentations. Hereafter, the 75 most similar titles will be referred to as a title's neighbors.

Once we knew what percentage of each title's neighbors were presentations, we then had to determine whether that percentage was high or low. Across the entire corpus of 12,199 presentations and articles, 43.7% were presentations. Thus, one way to determine whether a title is in a presentation-heavy area would be if a greater percentage of its neighbors were presentations than the corpus-wide 43.7% would predict. To make interpretation easier, we also expressed the proportions as a log-odds relative to the corpus' expected proportion. Expressing in log-odds leads to a value of 0 when they are the same (i.e., both are approximately 43.7%), positive when a title's neighbors contained more presentations than expected, and negative when a title's neighbors contained more articles than expected. To avoid mathematical errors when a title's neighbors had zero presentations or zero articles, we added 0.5 to each count before computing the log-odds (i.e., Haldane-Anscombe correction; Anscombe, 1956; Haldane, 1956). Each log-odds came with a 95% confidence interval showing how precisely it was estimated (Woolf, 1955). A title's neighborhood was, thus, considered presentation-heavy when the lower bound of that interval was above 0. With 75 neighbors, that meant at least 42 were presentations (56%). We chose 75 neighbors because a neighborhood of that size gave a precise estimate for each title while staying close to the title's subject (see Figure 4 for a supporting analysis).

Quantifying Publication Support. The second characteristic of each title that we wanted to calculate was how many journal-article titles were close to it. A title surrounded mostly by presentations, with few articles nearby, would denote an area where behavior analysts present often and publish little. This allows us to judge each by its neighbors as compared to trying to group titles arbitrarily into groups (i.e., topic modeling). This approach was taken because titles gradually changed from one subject into the next with few natural boundaries between them.

To avoid assigning a label of "low" or "high" publication support arbitrarily, we anchored the cutoff to the data themselves. Specifically, we computed the cosine similarity between every pair of titles in the corpora (Appendix A) and took the value that only 1% of pairs exceed (.525). An article was considered closely related to a title when the two were at least that similar. In other words, a closely related article is more similar to a title than 99 of every 100 randomly chosen pairs of titles. An article was never counted as closely related to itself. This led the median title to have 26 closely related articles (interquartile range 7 to 73; see Appendix B for the supporting analysis).

Flagging Titles as Presentation-Heavy with Low Article Support

A title was flagged when both of the above conditions held. That is, a flagged title was one: (a) whose most similar titles were more often presentations than predicted by the corpus overall and (b) that has fewer than 10 closely related articles. We chose an absolute cut for the second condition so that few would mean the same thing for every title. The number 10 was a judgment and we report how the results change under varying cut points in the robustness checks below. Both measures are needed to answer the primary question motivating this analysis. For example, the presentation "Improving Social Functioning for Children With Autism" sat among titles that were 72% presentations, yet it had 466 closely related articles. The presentation "Opportunities and Issues in Practitioner Publication" sat among titles that were 73% presentations and had one closely related article. Only the second was flagged.

Describing Flagged Titles by Subject Area

Summarizing individually flagged articles is difficult across 12,199 titles. To describe what the flagged titles were about, we grouped the titles into 200 groups using *k*-means clustering (Lloyd, 1982) with a fixed random starting point (seed 0), keeping the best of 10 runs, and reported the percentage of each group's titles that were flagged as presentation-heavy and with low article support. These groups served only as a way of reporting the results. The evidence for flagged titles was determined for each title as described above, which does not depend on the group each title falls within. We also checked and report on whether the subject-level summary depended on the particular grouping (see *Robustness Checks* below).

To make the groups easier to interpret, we sought to describe each group in words. A group was defined by its cluster center and its titles (see step 6 of the interactive supplement). To get a description of each group, we first listed each group's most characteristic words and phrases (i.e., those common in that group's titles but uncommon in other groups' titles) using an algorithm called class-based TF-IDF (Grootendorst, 2022). We then generated a short label and a one-sentence definition for each group using Claude Opus 5.5 (Anthropic, n.d.). For each group, the model was passed the 20 titles closest to the group's center as input as well as the three most similar neighboring groups. The input prompt asked to return as output words describing what the group covered and what separated it from its neighbors. The prompts, model version, and responses are archived with the analysis code and every label was reviewed. See step 6 of the interactive supplement for a visualization of how this step worked.

Results

Of the 12,199 titles, 4,939 (40%) sat in presentation-heavy neighborhoods, and 3,685 (30%) had fewer than 10 closely related articles. The two conditions were nearly unrelated with Spearman correlation between a title's presentation proportion and its number of closely related articles = .04. That is, knowing that presentations are common around a title was not indicative of how much closely related articles had been published. A total of 1,552 titles (12.7%) met both conditions and were flagged. These included 1,050 of the 5,327 presentations (19.7%) and 502 of the 6,872 articles (7.3%). The flagged articles are published work that sat in the same sparse, presentation-heavy locations. Figure 2 shows all 12,199 titles by the two measures and with the flagged titles in the lower-right region (black markers).

What the Flagged Titles Are About

Flagged titles were not spread evenly across subject areas. Of the 122 groups with at least 40 titles, the median group had 7% of its titles flagged, and 33 groups had none. Twenty groups had at least a quarter of their titles flagged which is about twice the overall corpus rate. Table 2 lists these twenty groups alongside the plain-language description of each. The highest percentage of flagged titles were in the groups labeled *Student and Early-Career Professional Development* (84% of 101 titles flagged), *Collaboration and Bridging Research with Practice* (80% of 151), *Applied Behavior Analysis as a Profession* (74% of 110), *Clinical Service Delivery and Care Systems* (70% of 92), and *Behavior Analysis in School Systems* (59% of 117). The list continues through groups on climate change and proenvironmental behavior, sexual behavior, behavioral economics, racism and criminal justice, professional ethics, behavioral pharmacology, the dissemination of behavior science, and staff training. It also includes basic and translational groups such as *Operant Choice* and *Reinforcement in Rodents* (43%) and *Schedule-Induced Drinking and Alcohol Seeking* (27%).

At the other end, no titles were flagged in groups such as *Stimulus Preference Assessment Methods* (23 presentations against 71 articles), *Functional Analysis Procedural Variations and Validity* (29 against 58), *Differential Reinforcement and Treatment Relapse* (14 against 51), *Verbal Behavior and Intraverbals in Autism* (21 against 79), and *Training Caregivers, Staff, and Autistic Adults* (48 against 74).

Counting within groups and looking at each title's neighbors did not always lead to similar results. For example, the group labeled *School Practice and Teaching Behavior Analysis* contained 101 presentations and 26 articles, a ratio that might suggest less published literature. All of its titles sat in presentation-heavy neighborhoods, yet only 2% were flagged because the median title in the group had 121 closely related articles. Most of those articles fell just outside the group's boundary. Similarly, the group labeled *Behavior Analysis's Role in Autism Treatment* contained 53 presentations and 29 articles but had no flagged titles, with a median of 229 closely related articles per title. Thus, a count within a group depended on where its group boundary was drawn.

Figure 3 shows where the groups sat relative to one another when the 384 dimensional space was collapsed down to two dimensions. To do this, we computed the cosine distance (1 minus cosine similarity) between every pair of group centers and applied metric multidimensional scaling fit by the SMACOF algorithm (de Leeuw & Mair, 2009; Torgerson, 1952), using scikit-learn 1.8 with eight random starts of up to 900 iterations each and keeping the solution with the lowest stress. In less technical terms, the algorithm placed each group on a flat page so that the distances between groups on the page match their distances in the full 384-dimensional space as closely as possible. A flat page is unable to preserve every distance perfectly, but the distances in Figure 3 were correlated with the true distances in 384-dimensional space at $r = .77$. Thus, Figure 3 shows the broad layout of the groups but not exact distances. Each group is shaded by the proportion of its titles that were flagged. The darkest groups sit together in the upper right of the map, among groups about the profession, its training, and its practice. A few darker groups sit in the lower half, including those on operant choice in rodents, behavioral economics, behavioral pharmacology, and sexual behavior.

Robustness Checks

We conducted a series of robustness checks to determine how the obtained results may have been influenced by the analytic choices made. The first concerned the two cutoffs. For neighbors flagged as presentations, we conducted a systematic sweep using 50 to 100 neighbors to quantify a title as presentation-heavy. And for similar articles, we conducted a sweep of the closely related article cutoff from fewer than 1 (i.e., none) to fewer than 20. Figure 4 shows the results of these sweeps. For example, using 50 instead of 75 neighbors while keeping the cutoff at fewer than 10 articles flagged 12.3% of titles (19.2% of presentations) compared to 12.7% of titles (19.7% of presentations) in the main analysis. As is readily viewable in Figure 4, changing the number of neighbors had little impact on titles, presentations, and articles flagged (they are all relatively flat across the "neighbors" dimension). However, changing the number of articles required to indicate a "low" count of similar published articles had a larger influence. Looser cutoffs

flagged more titles, presentations, and articles, though each additional article added less than the one before (e.g., 3.0%, 12.7%, and 17.4% of titles at cutoffs of fewer than 1, 10, and 20 articles).

The second robustness check concerned whether flagged titles would reappear in a replication. Finding a title's nearest neighbors involves no random starting point, so the same data always produce the same flags. A replication may, however, draw a somewhat different set of titles. We therefore removed a random 20% of titles, recomputed every neighborhood, the closely related cut, and every count on the remaining 80%, and repeated this 10 times. Because 20% of the articles were removed, the cut was also scaled to fewer than 8 articles. A median of 88% of the titles flagged in the full data remained flagged (median Jaccard index = .81; the number of titles flagged in both, divided by the number flagged in either).

The third check concerned writing style. Conference titles might be phrased differently from article titles (e.g., "Recent advances in . . ." against "The effects of . . ."). If so, presentations could sit near one another because of the intraverbals used rather than the behavior analytic specific topic. To conduct this check, we removed 47 leading phrases of this kind from the 1,911 titles containing one. We then embedded the shortened titles again, and recomputed every neighborhood. The presentation proportion around each title correlated at .99 with the original, and 97% of the titles in presentation-heavy neighborhoods remained in presentation-heavy neighborhoods.

The final check concerned the subject-level summary in Table 2 and Figure 3. If that summary depended on where the group boundaries fell, it would change when the groups were redrawn. We redrew the groups from seven other random starting points and with 100 and 300 groups. For every title, we then compared the proportion of flagged titles in the group it fell in under the original and the redrawn groupings. These values correlated between .76 and .83 (Spearman), suggesting that the subject-level picture held across groupings. The proportion reported for any single group should, however, be read as approximate.

Discussion

Behavior analysts present on some subjects more than they publish on them. This may occur for a variety of reasons such as frontier research is still working toward methods and findings that are rigorous and replicable enough to submit to peer review; or, perhaps, because professional contingencies do not support publishing relative to presenting. We sought to determine the extent to which topics presented at conferences between 2016-2026 also have similar peer-reviewed publications. Across 14 journals and three conference programs, 19.7% of presentations sat in reliably presentation-heavy neighborhoods and had fewer than 10 closely related articles. Those titles concentrate in work about the profession itself, its training, and its practice (e.g., professional development, collaboration between researchers and practitioners, clinical service delivery, and school systems), with smaller concentrations in some basic and translational subjects. The data describe subjects. They evaluate no individual speaker, conference, or journal.

The subject-level result is the structural counterpart to the speaker-level findings that prompted it (Ayres Boggs et al., 2026; Kranak & Onofrio, 2025; Rapp et al., 2025). When 14 of 17 speakers at a single conference have no topic-specific publications (Rapp et al., 2025), or when 31% of speakers across state or regional conferences do (Ayres Boggs et al., 2026), part of what those counts may reflect is that the topics being presented on have few publications for anyone to have authored. Relatedly, where a presentation sits in a presentation-heavy neighborhood and has fewer than 10 closely related articles, the pool of practitioners who could point to publications on that subject is small before any speaker is chosen. It is important to be careful with that inference because the speaker studies counted topic-specific publications across all subjects rather than these in particular. Whether zero-publication speakers concentrate among the flagged subjects is a testable prediction for work that joins speaker-level and title-level data. It is not something these data establish.

Comparing each title with its neighbors also changed which subjects had less publication support. Counting presentations and articles within a fixed group treats the group's boundary as real. The school-practice group described above held about four presentations for every article, a pattern that would ordinarily mark a scant published literature base, yet its titles sat close to a large body of educational research that fell into neighboring groups. In a field whose titles form a continuum, a count within a boundary depends on where the boundary is drawn. Future bibliometric work in behavior analysis may therefore benefit from measures that do not require boundaries at all.

There seems to be an additional qualification worth making explicit. Some subjects of importance to behavior analysts are sought out for discussion at conferences and are genuinely hard to study with the within-subject designs most readily published in behavior-analytic journals. For example, consider staff training and supervision, whose outcomes of interest (e.g., supervisee competencies across a cohort, turnover, the fidelity of supervision across many dyads) accrue across people and are often better evaluated with group or longitudinal designs than with a reversal on one supervisee. As another example, consider service delivery where the questions are program-evaluation and

implementation science questions before they are single-case ones (e.g., whether an organization's operational model improves access or outcomes across its clients). To the extent that behavior analysts are seldom trained in group, qualitative, implementation, or program-evaluation methods, the skills needed to build these literatures may themselves be a rate-limiting factor. Critchfield and Reed (2017) made a related point a decade ago, arguing that the seven-dimension definition of good applied work (Baer et al., 1968) can function as a bottleneck that keeps socially important but methodologically awkward problems underrepresented in the literature.

Limitations

There are several important limitations with these data and analysis. First, conference records were title-level only, so each title's neighbors are judged from a dozen or so words, and a vague or playful title (e.g., "Take Your Medicine!") may have few closely related articles simply because its wording says little about its subject. Second, conference coverage spanned 2016-2026 from three specific conferences and so is biased in both regards. ABAI also contributed 83% of the *presented* corpus so the pooled results may describe that convention more than it describes the other two. And, recent subjects might be under-published partly because publication lags presentation. Third, the *presented* corpus pooled event types that are not necessarily equivalent to one another. A symposium slot and an invited address differ in how much content they represent and in who is likely to attend. Weighting these differentially was difficult to justify but could influence the results obtained. Lastly, a count of closely related articles is a coarse proxy for a research base and speaks to no individual's expertise, given that absence of publication is not absence of evidence. Even across 14 journals, some relevant work lives in adjacent literatures (e.g., general psychology, education, or medicine) that was not present in the current *published* corpus, so the counts of closely related articles are lower bounds.

Two additional limitations deserve their own paragraph. The first concerns the analytic choices. The number of neighbors (75), the definition of closely related (the 99th percentile of similarity), and the cut of fewer than 10 articles were analytic choices and other defensible values flag somewhat different titles (Figure 4). The checks reported above suggest that the subjects leading the list changed little across those choices, while the number of flagged titles depends on how strictly "few" is defined. Neighborhoods also overlap, so the 95% intervals describe each title's own neighborhood and are not independent tests across titles. Table 2 should be read as a broad description, built from groups drawn for reporting, of where the field's presented and published work diverge, and not as a definitive enumeration of subjects.

The second concerns what the comparison supports. We described two distributions. We did not observe why they differ and several mechanisms could produce the same pattern. For example, a subject might be presented often because it is new, because it is practice-facing, because it resists the experimental designs that journals favor, or because it draws attendees. Each of these implies something different about what the field should do, and these data cannot distinguish among them.

Moving forward

The subjects that behavior analysts present on most, relative to what they publish, are a describable set concentrated in the profession's practice, its workforce, and its training. Future research might join speaker-level and title-level data to test whether zero-publication speakers concentrate in these subjects, extend the comparison to abstracts rather than titles so that each neighborhood is judged on more than a title's wording, and examine whether the gap around a given subject narrows in the years after it appears. For the subjects themselves, the more useful next step may be methodological. Several of the subjects identified ask questions that accrue across people and settings, and the field's published base for them is likely to grow only as behavior analysts contact and publish the designs suited to such questions. The same studies that would answer the call for better continuing-education content would also, subject by subject, give the field more to say.

References

Anscombe, F. J. (1956). On estimating binomial response relations. *Biometrika, 43*(3–4), 461–464. https://doi.org/10.1093/biomet/43.3-4.461

Anthropic. (n.d.). *Claude Opus 5.5* [Large language model]. https://www.anthropic.com

Ayres Boggs, M., Rapp, J. T., Cook, J. L., Richling, S. M., & Mohamed, R. (2026). Are keynote and invited speakers at state behavior analytic conferences experts on their presentation topics? *Behavior Analysis in Practice, 19*(3), 877–888. https://doi.org/10.1007/s40617-025-01051-2

Baer, D. M., Wolf, M. M., & Risley, T. R. (1968). Some current dimensions of applied behavior analysis. *Journal of Applied Behavior Analysis, 1*(1), 91–97. https://doi.org/10.1901/jaba.1968.1-91

Behavior Analyst Certification Board. (2025). *BCBA handbook*. https://www.bacb.com/bcba-handbook/

Behavior Analyst Certification Board. (2026). *ACE provider handbook: Effective July 1, 2026*. https://www.bacb.com/wp-content/ACE-Provider-Handbook-2026

Critchfield, T. S., & Reed, D. D. (2017). The fuzzy concept of applied behavior analysis research. *The Behavior Analyst, 40*(1), 123–159. https://doi.org/10.1007/s40614-017-0093-x

Gentzkow, M., Kelly, B., & Taddy, M. (2019). Text as data. *Journal of Economic Literature, 57*(3), 535–574. https://doi.org/10.1257/jel.20181020

Glenn, S. S. (1988). Contingencies and metacontingencies: Toward a synthesis of behavior analysis and cultural materialism. *The Behavior Analyst, 11*(2), 161–179. https://doi.org/10.1007/BF03392470

Grimmer, J., & Stewart, B. M. (2013). Text as data: The promise and pitfalls of automatic content analysis methods for political texts. *Political Analysis, 21*(3), 267–297. https://doi.org/10.1093/pan/mps028

Grootendorst, M. (2022). *BERTopic: Neural topic modeling with a class-based TF-IDF procedure* (arXiv:2203.05794). arXiv. https://doi.org/10.48550/arXiv.2203.05794

Haldane, J. B. S. (1956). The estimation and significance of the logarithm of a ratio of frequencies. *Annals of Human Genetics, 20*(4), 309–311. https://doi.org/10.1111/j.1469-1809.1955.tb01285.x

Kranak, M. P., & Onofrio, O. (2025). Selecting speakers for behavior-analytic conferences: Policy review and recommendations. *Behavior and Social Issues, 34*(1), 110–133. https://doi.org/10.1007/s42822-024-00193-2

de Leeuw, J., & Mair, P. (2009). Multidimensional scaling using majorization: SMACOF in R. *Journal of Statistical Software, 31*(3), 1–30. https://doi.org/10.18637/jss.v031.i03

Lloyd, S. (1982). Least squares quantization in PCM. *IEEE Transactions on Information Theory, 28*(2), 129–137. https://doi.org/10.1109/TIT.1982.1056489

Mikolov, T., Sutskever, I., Chen, K., Corrado, G. S., & Dean, J. (2013). Distributed representations of words and phrases and their compositionality. *Advances in Neural Information Processing Systems, 26*. https://proceedings.neurips.cc/paper_files/paper/2013/hash/9aa42b31882ec039965f3c4923ce901b-Abstract.html

Mohamed, R., Cook, J. L., Ayres Boggs, M., & Rapp, J. T. (2026). Analysis of two metrics for keynote and invited speaker expertise: A replication and extension in Canada. *Behavioral Interventions, 41*(2), Article e70104. https://doi.org/10.1002/bin.70104

Rapp, J. T., Ayres Boggs, M., Cook, J. L., Richling, S. M., & Hatch, R. W. (2025). The question of expertise and scientific rigor at ABA conferences for CEUs: A case example and analysis of the SexABA 2025 conference. *Behavioral Interventions, 40*(3), Article e70019. https://doi.org/10.1002/bin.70019

Reimers, N., & Gurevych, I. (2019). Sentence-BERT: Sentence embeddings using Siamese BERT-networks. In K. Inui, J. Jiang, V. Ng, & X. Wan (Eds.), *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)* (pp. 3982–3992). Association for Computational Linguistics. https://doi.org/10.18653/v1/D19-1410

Shadish, W. R., Zelinsky, N. A. M., Vevea, J. L., & Kratochwill, T. R. (2016). A survey of publication practices of single-case design researchers when treatments have small or large effects. *Journal of Applied Behavior Analysis, 49*(3), 656–673. https://doi.org/10.1002/jaba.308

Torgerson, W. S. (1952). Multidimensional scaling: I. Theory and method. *Psychometrika, 17*(4), 401–419. https://doi.org/10.1007/BF02288916

Wang, W., Wei, F., Dong, L., Bao, H., Yang, N., & Zhou, M. (2020). MiniLM: Deep self-attention distillation for task-agnostic compression of pre-trained transformers. *Advances in Neural Information Processing Systems, 33*, 5776–5788. https://proceedings.neurips.cc/paper/2020/hash/3f5ee243547dee91fbd053c1c4a845aa-Abstract.html

Woolf, B. (1955). On estimating the relation between blood group and disease. *Annals of Human Genetics, 19*(4), 251–253. https://doi.org/10.1111/j.1469-1809.1955.tb01348.x

Tables

Table 1. Pseudocode for transforming titles into vectors of numbers.

```
import all-MiniLM-L6-v2 as model
import title_dataset
vectors_list = empty list
for each title in title_dataset:
vector = model(title) # returns 384 number list
append vector to vectors_list
record title_id alongside vector
save vectors
```

Table 2. Subject-area groups in which at least a quarter of titles were flagged, ranked by the share flagged. Pres. = presentations; Pub. = articles.

Group	What it covers	Pres.	Pub.	Flagged titles (%)
Student and Early-Career Professional Development	Guidance on building careers in behavior analysis, including academic job searches, graduate training and coursework, supervision relationships, conference navigation, and expanding practice into new areas after graduation.	76	25	85 (84%)
Collaboration and Bridging Research With Practice	Work on interdisciplinary and interprofessional collaboration, community-engaged partnerships, and practitioners conducting or integrating research, including equity in research and closing gaps between researchers and applied practitioners.	96	55	121 (80%)
Applied Behavior Analysis as a Profession	Work on applied behavior analysis as a field and profession: regulation, credentialing, advocacy, training programs, international growth, practitioner ethics and skills, and how services are delivered across settings.	79	31	81 (74%)
Clinical Service Delivery and Care Systems	Work on how clinical behavioral services are organized and delivered: data-based decisions, clinician training systems, consent and assent, insurance and legal requirements, quality metrics, integrated healthcare collaboration, and research within patient care.	57	35	64 (70%)
Behavior Analysis in School Systems	Applying behavior analysis in elementary, secondary, and alternative schools, including group contingencies, school-based interventions, restraint reduction, school reintegration, inclusion models, practicum development, and system-level concerns such as equity and educational reform.	66	51	69 (59%)
Climate Change and Proenvironmental Behavior	Applying behavior analysis to environmental sustainability, including climate change, conservation of water and shared resources, reducing emissions, community interventions, and calls for the field to act on ecological problems.	42	32	41 (55%)
Sexual Behavior Research and Intervention	Behavior-analytic assessment, treatment, and study of sexual behavior, including inappropriate or illegal sexual behavior, arousal and partner-preference assessment, sexual risk discounting, sex education, and ethics of sexuality-related interventions.	40	40	40 (50%)
Behavior Analysis as a Science	Work on the nature, culture, methods, and dissemination of science and scientists' behavior, including scientific practice, advocacy, free will and determinism, subjectivity, and tributes to contributing scientists.	20	26	20 (43%)
Operant choice and reinforcement in rodents	Laboratory studies of rats and mice examining operant responding, schedules of reinforcement, delay discounting and intertemporal choice, foraging, cooperation, social preference, aversive consequences, and relapse phenomena such as reinstatement and renewal.	64	29	40 (43%)
Field history, future directions, and reflections	Broad reflective pieces on behavior analysis's past and future, including historical figures and anniversaries, international trends, book reviews, commentaries, and big-picture calls for systemic or societal change.	15	37	22 (42%)
Behavioral Economics Applications and Policy	Broad overviews and applications of behavioral economics to research, practice, public policy, health, sustainability, poverty, consumer behavior, and the economics of the behavior-analytic profession, such as fees and pay equity.	47	25	28 (39%)
Racism, Policing, and Criminal Justice	Behavioral and systems analyses of racism, bias, oppression, and violence, especially in policing and criminal justice, including anti-racist interventions, restorative justice, and calls for behavior analysis to become anti-racist.	53	53	36 (34%)
Professional Ethics and Ethical Decision-Making	Work on ethical conduct in behavior-analytic practice and research, including interpreting and applying the professional ethics code, models of ethical decision-making, ethics training, and ethical dilemmas such as punishment.	94	53	48 (33%)

Group	What it covers	Pres.	Pub.	Flagged titles (%)
Behavioral Pharmacology and Psychotropic Medication	Work on behavioral pharmacology as a discipline, covering its methods, training, and regulatory roles, and on how psychotropic and prescription medications affect problem behavior, assessment, and treatment in applied behavior analysis.	32	14	15 (33%)
General Behavior Change Principles and Procedures	Broad, mixed work on core behavior-change concepts and methods, including shaping, chaining, defining target behavior, motivating operations, behavior plans, functional analysis of problem behavior, and goal setting.	28	25	17 (32%)
Disseminating and Expanding Behavior Science	Work on spreading behavior science beyond its usual settings through public communication, plain language, social media, entrepreneurship, technology, artificial intelligence, community psychology and public policy, broadening its reach and impact.	47	48	29 (31%)
Refining Skill Acquisition Teaching Procedures	Research on making skill acquisition programs more effective and efficient, examining antecedent and consequence variables such as differential reinforcement, feedback timing, instructive feedback, mastery criteria and learning arrangements.	73	60	37 (28%)
Schedule-Induced Drinking and Alcohol Seeking	Research on schedule-induced (adjunctive) drinking in rats, along with alcohol and ethanol self-administration, choice, demand, resurgence of alcohol seeking, and alcohol-related behavior in humans.	27	18	12 (27%)
Staff Training and Performance Management	Research on training, supervising, and managing direct-care and human service staff, including pyramidal and remote training, performance feedback, self-monitoring, performance diagnostics, and reactivity in implementer fidelity, in residential and applied settings.	52	47	25 (25%)
Behavioral Accounts of Mind and Thought	Conceptual and experimental behavior-analytic treatments of mentalistic phenomena such as theory of mind, consciousness, thinking, self-talk, perspective taking, implicit cognition, and self-referential thoughts.	30	34	16 (25%)

Figures

Figure 1. Demonstration for how similarity is measured between two embedded titles.

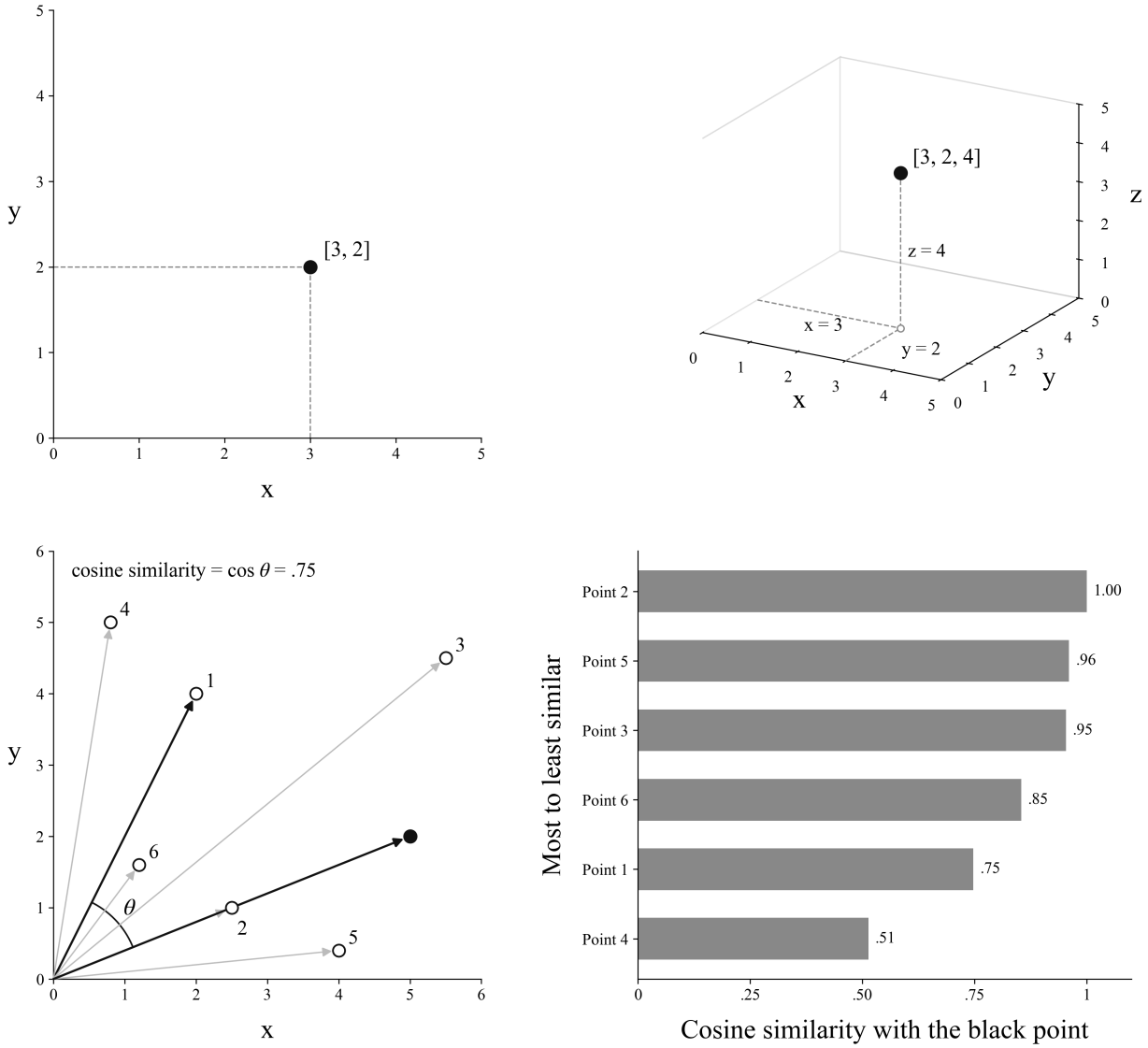
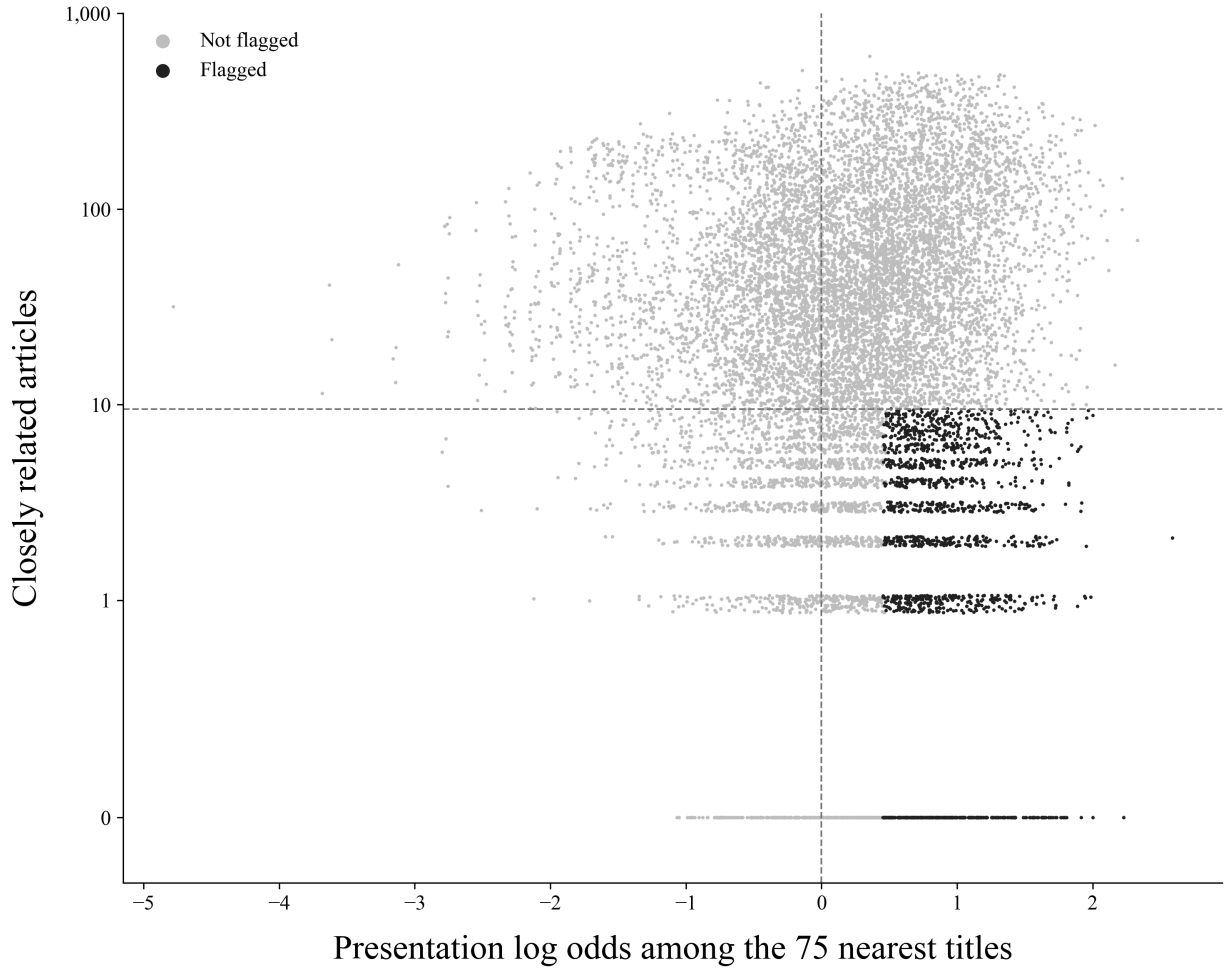
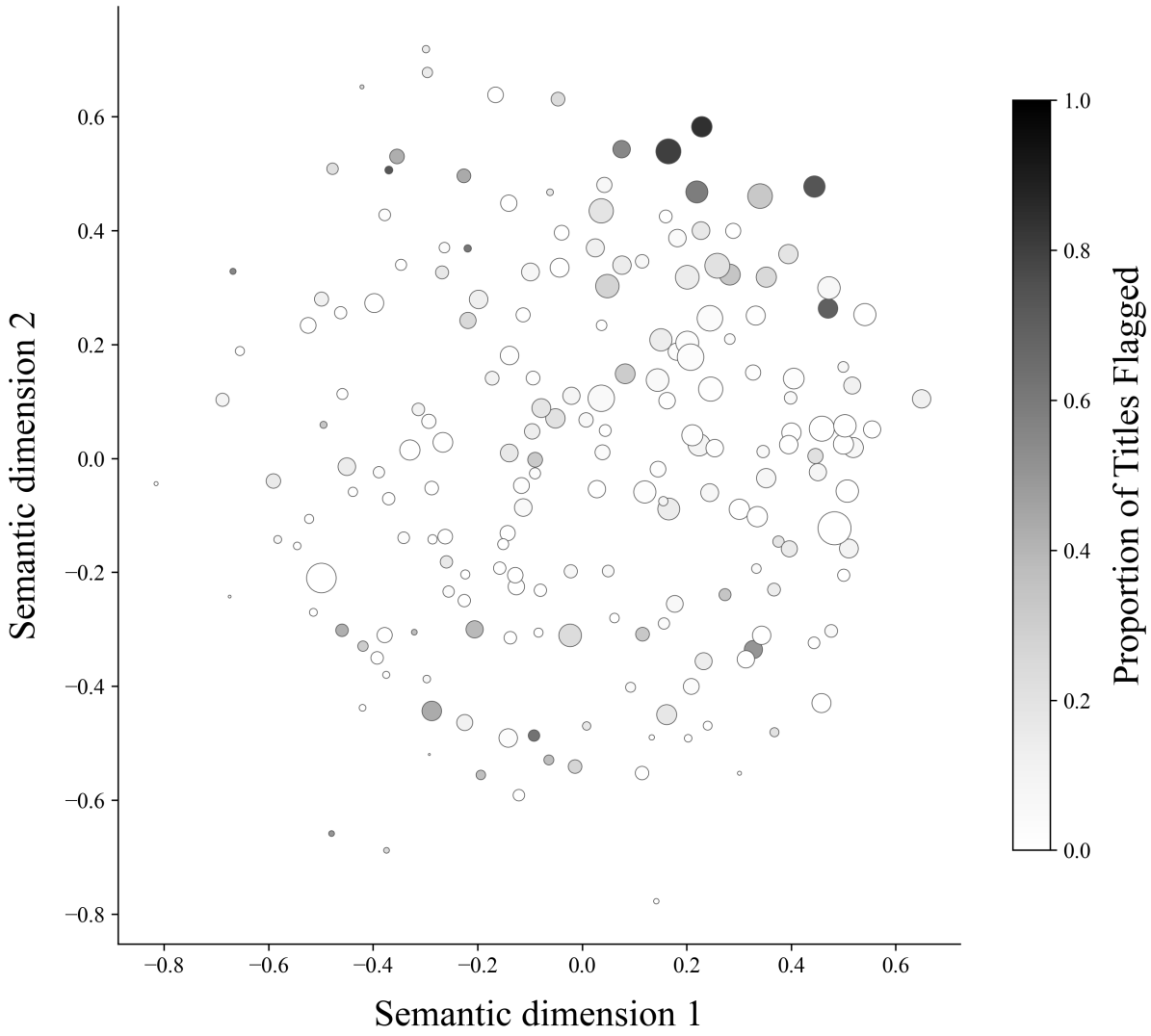


Figure 2. Every title (N = 12,199) placed by the log-odds of presentation proportion among its 75 most similar titles (x-axis) and its number of closely related articles (y-axis).



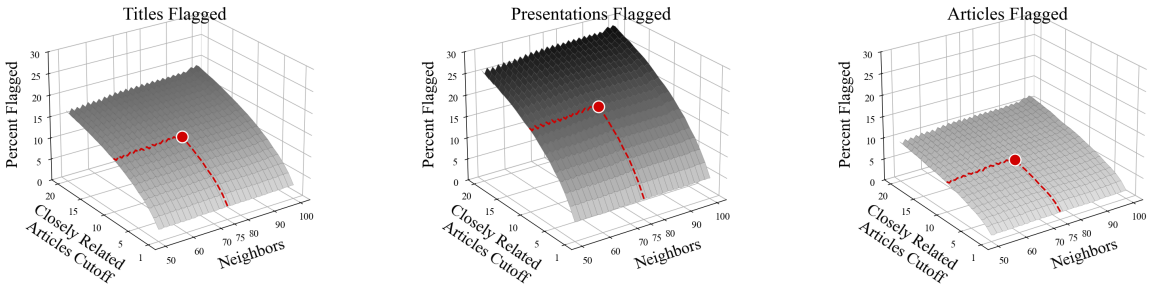
Note: Black points are the 1,552 flagged titles having presentation-heavy neighbors and fewer than 10 articles. The vertical dashed line marks a log-odds of zero, where presentations around a title match the corpus-wide percentage of 43.7%. The horizontal dashed line marks the cut of fewer than 10 closely related articles. A title is flagged when the lower bound of its log-odds interval is above zero, so flagged titles begin to the right of the vertical line. Points are jittered slightly for display.

Figure 3. Map of the subject-area groups, shaded by the share of each group's titles that were flagged.



Note: Each point is a group of titles, placed by metric multidimensional scaling so that groups with similarly worded titles sit closer together (correlation between plotted and full-space distances = .77). Point area is proportional to the number of titles in the group, and darker shading means a larger share of the group's titles were flagged. Groups are a reporting device; each title's flag does not depend on its group.

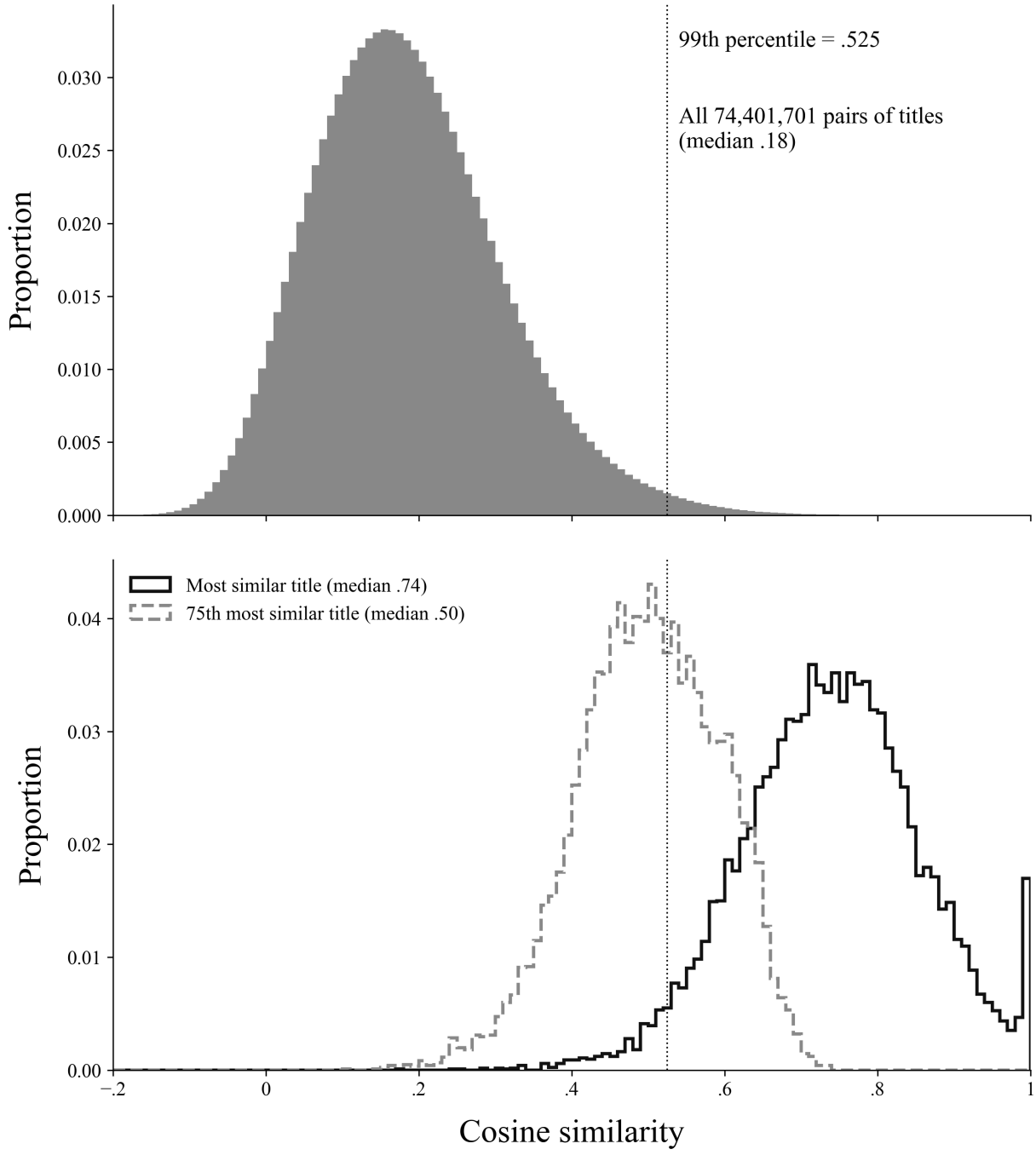
Figure 4. Influence of cutoff metrics on titles flagged. The percentages flagged were 12.7% (titles), 19.7% (presentations), and 7.3% (articles). The red dot is the point used in the main analyses.



Red: The Definition Used in the Main Analysis (75 Neighbors, Fewer Than 10 Closely Related Articles), With Dashed Lines Along the Surface at Each Value. Same Vertical Scale in All Panels.

Appendix A

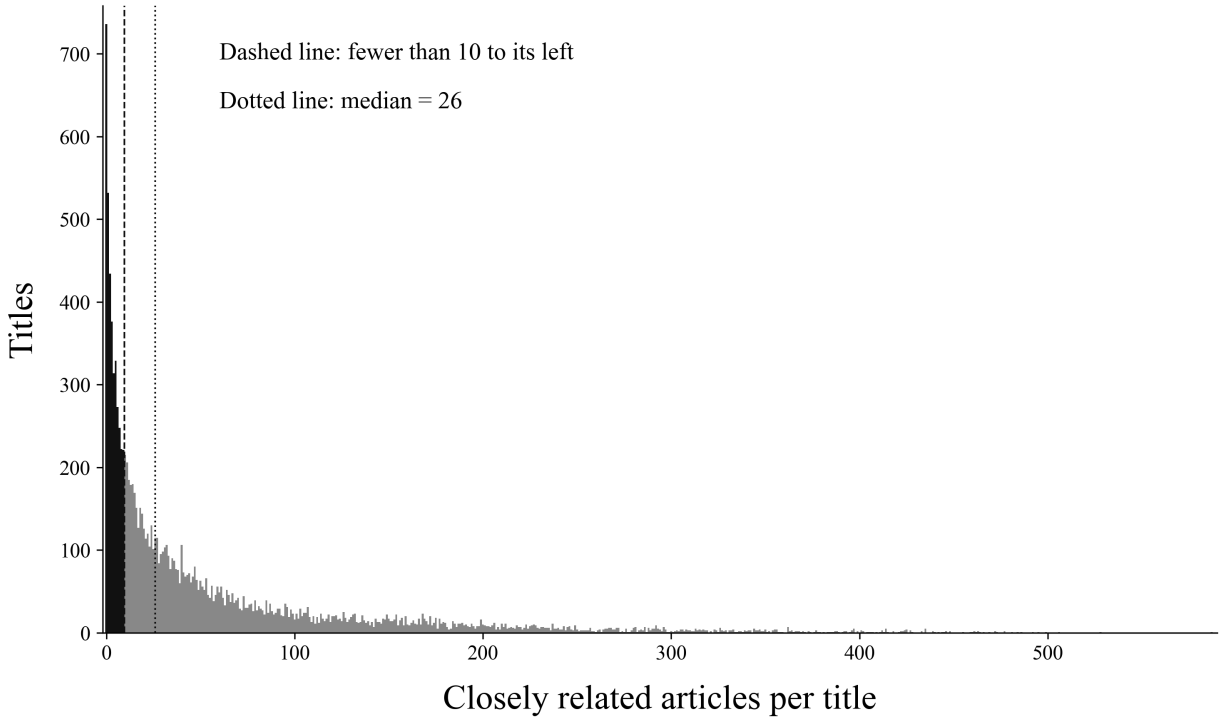
Figure A1. Distributions of cosine similarity between titles in the corpus.



Note: The top panel shows the distribution of cosine similarity for all 74,401,701 distinct pairs among the 12,199 titles. The median cosine similarity is .18. The bottom panel shows the distribution of each title's similarity to its most similar title (solid line; median .74) and to its 75th most similar title (dashed line; median .50), the outer edge of the neighborhood used to define presentation-heavy areas. The dotted line in both panels marks .525, the 99th percentile of all pairs; an article with the log odds 95% confidence interval at or above this similarity to a title counted as closely related to it.

Appendix B

Figure B1. Number of closely related articles per title.



Note: An article was closely related to a title when their cosine similarity was at least .525, the 99th percentile of similarity over all distinct pairs of titles (Appendix A). Black bars are the 3,685 titles (30%) with fewer than 10 closely related articles. The median title had 26 closely related articles (interquartile range 7 to 73), and 736 titles (6.0%) had none.

Table B1. Closely Related Articles per Title When the Line Is Drawn at Other Percentiles

Percentile of All pairs	Similarity Line	Median (IQR)	Titles with None	Titles with Fewer Than 10
95th	.406	202 (71–467)	0.7%	5.2%
97.5th	.460	80 (26–212)	2.0%	12.5%
99th	*.525*	*26 (7–73)*	*6.0%*	*30.2%*
99.5th	.570	11 (2–34)	12.0%	47.1%
99.9th	.662	2 (0–6)	35.3%	81.7%

Note: The analysis uses the 99th percentile (italics). IQR = interquartile range. Lower percentiles draw the line at a lower similarity, so more articles count as closely related and fewer titles fall below 10; higher percentiles do the reverse.